MIL-A-16749E(SHIPS)
15 May 1963
SUPERSEDING
MIL-A-16749D(SHIPS)
13 April 1956

MILITARY SPECIFICATION

AMPLIFIER, AUDIO FREQUENCY, SOUND POWERED TELEPHONE SYSTEM

1. SCOPE

1.1 This specification covers a type AM 2210( )/WTC amplifier used to amplify audio frequency signals sufficiently to drive a group of sound powered telephones and loudspeakers at a shipboard location having a high ambient noise level.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-I-983 - Interior Communication Equipment; Naval Shipboard, Basic Design Requirements for.
MIL-P-15024 - Plates, Identification - Information and Marking for Identification of Electrical, Electronic and Mechanical Equipment.
MIL-M-15071 - Manuals, Equipment and Systems.
MIL-E-17362 - Electronic Repair Parts Requirements, Procedures for Provisioning Technical Documentation and Stock Numbering.
MIL-E-17555 - Electronic and Electrical Equipment and Associated Repair Parts; Preparation for Delivery of.
MIL-E-21981 - Electronics Type Designations, Identification Plates and Markings; Requirements for.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules

(Application for copies should be addressed to Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, New York.)

3. REQUIREMENTS

3.1 Preproduction sample. - Prior to beginning production a sample shall be tested as specified in 4.2.1 (see 6.3.1).

3.2 Description. -

3.2.1 The sound powered telephone amplifiers shall be designed to provide amplified one-way communication in two-way sound powered telephone communication system. The amplifier shall be designed to reproduce voice signals at a level such that the message will be clear and understandable through the high noise levels at a gun position or in a machinery room. The two-way sound powered telephone circuit will consist of one or more sound powered telephone headsets or handsets at a remote station, connected via

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the sound powered telephone amplifier to a local group of one to six sound powered telephone headsets and one or two loudspeakers.

3.2.2 The type designation for the amplifier, indicated in 1.1, contains parenthesis which are to be replaced by a letter indicating a particular make or model that differs from previous production. Nomenclature for a production order shall be requested as required by MIL-E-21981.

3.3 General features. - The equipment shall be in accordance with the following paragraphs of MIL-I-983 in addition to the requirements specified herein. (Whenever a requirement of MIL-I-983 conflicts with a requirement of this specification, the requirement of this specification shall govern).

General requirements
Definitions
Materials, general
Substitution of (equal or superior) materials or parts
Fungus-inert materials
Unacceptable materials
Acceptable materials
Flammable materials
Arc-resistant materials
Toxic materials
Wood
Metals
Aluminum
Magnesium
Iron and steel
Nonferrous material (except aluminum)
Zinc
Springs (material)
Other metals
Plastics
Ceramics
Impregnating, embedding and encapsulating compounds
Painting
Protection against corrosion
Bolts, machine screws, studs and nuts
Locking devices
Washers
Parts- electrical - general
Use of nonstandard parts
Requirements for semiconductor devices
Electron tube or capacitor sockets
Capacitors (see 3.5.14)
Variable resistors
Transformers
Relays
Electrical tapes
Batteries
Switches
Indicator lights and lampholders (see 3.5.10.2)
Fuses (see 3.5.11)
Fuseholders and fuse clips (see 3.5.11)
Metallic rectifiers
Printed wiring and circuits
Enclosures - general
Enclosure - accessibility
Enclosure - degree of (dripproof)
Enclosure - Mounting (bulkhead)
Cable entrance
Ventilation

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Threaded devices
Rounded corners and edges
Internal subassembly protection
Drilled and tapped holes
Welding
Temperature and humidity
Shock, vibration and inclination
Power supply tolerances (normal power source - 115 volts, 60 cycles)
Personnel protection
Shielding and radio frequency noise reduction
Ground potential and grounding
Soldering
Electrical parts mounting
Internal subassembly connection (Test cable required see 3.5.12 and 3.5.15)
Terminal boards connectors and terminals
Wiring
Electrical insulation
Drawings - general
Drawings - preliminary
Drawings - working
Interchangeability and standardization
Manuals (see 3.7)
Repair parts (electronic)
Designation and marking (Serial numbers and stock numbers are required)
Workmanship and general examination

3.4 Special design features. - The sound powered telephone amplifier shall provide the following features:

3.4.1 Under normal conditions, with the amplifier turned on, incoming signals from the remote sound powered telephone station shall be amplified and transmitted to the sound powered telephone headsets and the loudspeakers at the local station. The amplifier shall incorporate a sensitive switching means which shall operate to reduce the amplifier output to zero and to connect the headsets at the local station to the remote station when the talk switch of any of the local station headsets is operated. The sound powered telephone headsets will have d.c. blocking capacitors in their receiver circuits. The d.c. resistance of the microphone unit will be approximately 4.6 ohms. The sensitive switching circuit current, which flows through the microphone unit and sensitive relay coil when the talk switch is operated, shall not exceed 3.5 milliamperes. The voltage of this switching circuit shall not exceed 4 volts d.c. The sensitive switching circuit shall function properly with a minimum of 40 ohms resistance connected in series with the microphone unit.

3.4.2 In case of amplifier power supply failure or failure of the amplifier's sensitive switching means, two-way sound powered telephone communication shall be maintained between the local and remote stations.

3.4.3 When the amplifier is switched off, two-way sound powered telephone commumication shall be maintained between the local and remote stations (see 3.5.10.1).

3.4.4 The amplifier circuit shall include means of compensating for the sharply peaked response characteristics of sound powered telephone microphone (see 3.5.5).

3.5 Electrical features. -

3.5.1 Input impedance. - The impedance of the input transformer with the amplifier energized shall be 800 ± 200 ohms, resistive at any frequency between 400 and 2,500 cycles per second (c.p.s.).

3.5.2 Output voltage. - Output terminals shall be provided for six sound powered telephone headsets and for one or two loudspeakers. Outputs of 832, 525, and 331 millivolts root mean square (r.m.s.) at the telephone terminals shall be obtained through the sensitive switching network, from suitable taps on the output transformer. Outputs of 70 volts r.m.s. for one loudspeaker and 49 volts r.m.s. for two loudspeakers in parallel shall be obtained direct from the output transformer. The output voltages with tolerances of plus or minus 5 percent, shall be obtained when delivering output power into a resistive load of

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650 ohms across the 70-volt terminals or 325 ohms across the 49-volt terminals and 600 ohms across the output terminals for one headset.

3.5.3 Output power. - Rated output power shall be at least 7.5 watts, available at the output terminals for one loudspeaker.

3.5.4 Amplifier gain. -

3.5.4.1 Rated gain. - With a 1,000 c.p.s. sine-wave signal of 25 ± 2 millivolts r.m.s. applied to the amplifier input terminals, the amplifier shall deliver rated output power at rated voltage.

3.5.4.2 Gain control. - Gain control shall be provided, operable from the front of the amplifier, capable of reducing the output at least 24 decibels (db) from rated output in not less than five essentially equal steps. Operation of the gain control shall not introduce noise into the amplifier circuit.

3.5.5 Frequency response. - The output voltage at the output terminals of the loudspeaker shall be as shown on figure 1 over a frequency range of 200 to 4,000 c.p.s. when measured as specified in 4.4.3.

3.5.6 Distortion. - The total harmonic distortion of the output voltage at the output terminals of the loudspeakers shall not exceed 5 percent of the fundamental frequency when measured as specified in 4.4.4.

3.5.7 Noise and microphonics. - The self-generated voltage at the output terminals of the loudspeaker shall not exceed 350 millivolts (46 db below rated output voltage of 70 volts). The self-generated voltage at one pair of the sound powered telephone output terminals shall not exceed 3 millivolts. Both output voltages shall be measured as specified in 4.4.5.

3.5.8 Output voltage regulation. - A variation in the amplifier output load between full loan and no load shall not cause the output voltage, at the output terminals of the loudspeaker, to exceed 99 volts r.m.s (3 db above rated output voltage of 70 volts) when measured as specified in 4.4.6.

3.5.9 Stability. - The amplifier shall show no evidences of instability or oscillation when tested as specified in 4.4.7.

3.5.10 Controls. - The sound powered telephone amplifer shall have the following operating controls, mounted on the ront panel of the amplifier enclosure:

3.5.10.1 Power switch. - Operation of the power switch shall place the amplifier in a ready condition by connecting the available 115-volt, 60-cycle, single-phase power supply to the amplifier. In addition, another section of the switch shall connect the input signal to the amplifier and the proper amplifier output to the telephone output terminals when the switch is in the "on" position and shall connect the input signal direct to the telephone output terminals when the switch is in the "off" position.

3.5.10.2 Power indicator light. - The "power on" indicator light shall be provided with a red lens and shall have facility for mechanical dimming.

3.5.10.3 Volume control. - A knob shall be provided to reduce the amplifier gain in accordance with 3.5.4.2. The setting shall be indicated by a pointer on the knob and a scale on the panel.

3.5.11 Fuses and blown fuse indicators. - Fuses and fuseholders, in accordance with MIL-I-983, shall be provided in each side of the power supply, connected between the power switch and the power transformer. The current rating of the fuses shall be the next standard size above the current requirement of the amplifier.

3.5.12 Electrical connections. - All electrical connections to the amplifier chassis shall be made by means of plug-and-jack connectors. The front panel and chassis shall be removable as a unit from the case. Removal of the chassis shall automatically break all electrical connections between ship's power

[illegible] output transformer. Outputs of 70 volts r.m.s. for one loudspeaker and 49 volts r.m.s. for two loudspeakers in parallel shall be obtained direct from the output transformer. The output voltages with tolerances of plus or minus 5 percent, shall be obtained when delivering output power into a resistive load of

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3.5.13 Semi-conductor devices. - Electron tubes shall not be used. Semi-conductor devices shall be used and shall comply with the requirements of MIL-I-983.

3.5.14 Capacitors. - Electrolytic capacitors may be used. Their use shall be governed by the requirements of MIL-I-983.

3.5.15 Test cable and continuity plug. - One test cable assembly as specified in MIL-I-983 shall be required for each onboard repair part set. In addition, one plug shall be furnished as an accessory, wired to maintain communication between local and remote stations while the chassis is not connected to or inserted in the enclosure case (see 3.5.12).

3.6 Mechanical features. -

3.6.1 Enclosure. - The enclosure for the amplifier shall be of dripproof construction and designed for bulkhead mounting. The front of the enclosure shall consist of the mounting panel for the major parts.

3.6.2 Maximum weight and dimensions. - The amplifier shall not exceed 10 inches in width, 8 inches in height and 9 inches in depth including projection on the front panel. The weight of the amplifier shall not exceed 20 pounds.

3.6.3 Suitable handles shall be provided on the front panel to facilitate removal and replacement of the amplifier chassis.

3.6.4 Sufficient space shall be provided at the bottom of the amplifier enclosure to accommodate stuffing tubes to be furnished by the installing activity.

3.6.5 An instruction plate shall be mounted on one side of the case to explain briefly the method of operation of the unit.

3.6.6 A type A or B identification plate in accordance with MIL-P-15024 shall be mounted on the front panel. Equipment and component lines shall read, "Sound Powered Telephone Equipment" and "Sound Powered Telephone Amplifier". Serial numbers shall be carried forward consecutively for the same model amplifier. Federal stock numbers shall be included on the identification plate prior to shipment. The stock number shall be obtained in accordance with the procedures set forth in Appendix A of MIL-E-17362.

3.7 Manuals. - Manuals shall be in accordance with the requirements of MIL-I-983 and Type II of MIL-M-15071. Reproductions of the final working drawings shall be included in the manuals.

3.8 Equipment and maintenance repair parts. - Equipment and maintenance repair parts shall be furnished in accordance with MIL-E-17362. Equipment maintenance parts shall be type I but each set of equipment maintenance parts shall include parts in quantities based on the number per ten amplifiers instead of the number per equipment (see 3.5.15).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction inspection. - Preproduction inspection shall consist of the examination and tests specified in 4.2.1.

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4.2.1 Preproduction inspection shall consist of the examination and tests listed below, which shall be performed in general in the order listed:

Test	Requirement paragraph	Test paragraph
General examination	3.2	MIL-I-983
Operating	3.4	4.4.1
Supply line voltage and frequency	3.2	MIL-I-983
Input impedance	3.5.1	4.4.2
Output voltage	3.5.2	4.4.2
Output power	3.5.3	4.4.2
Amplifier gain	3.5.4	4.4.2
Frequency response	3.5.5	4.4.3
Distortion	3.5.6	4.4.4
Noise and microphonics	3.5.7	4.4.5
Transients	3.2	MIL-I-983
Output voltage regulation	3.5.8	4.4.6
Stability	3.5.9	4.4.7
Radio frequency noise reduction	3.2	MIL-I-983
Temperature and humidity	3.2	MIL-I-983
Enclosure	3.2	MIL-I-983
Shock and vibration (vital test required)	3.2	MIL-I-983

Test shall be monitored and test results certified by the Government. Test results shall be forwarded to the bureau or agency concerned for approval.

4.3 Sampling for quality conformance inspection. -

4.3.1 Inspection lot. - All equipment presented for delivery at one time shall be considered a lot. The lot may include the entire contract quantity or it may be the production of any convenient time period.

4.3.2 Group A and B. - All equipment covered on the contract or order shall be subjected to the group A and B examination and tests in the order listed. The results of each test shall be compared with specification requirements. In the event of failure to conform to this specification for any examination or test, the contractor shall correct the case of failure on future production units and repair the deficiency in all equipments produced on the contract or order.

Test	Requirement paragraph	Test paragraph
Group A tests		
General examination	3.2	MIL-I-983
Operating	3.4	4.4.1
Group B tests		
Supply line voltage and frequency variation	3.2	MIL-I-983
Input impedance	3.5.1	4.4.2
Output voltage	3.5.2	4.4.2
Output power	3.5.3	4.4.2
Amplifier gain	3.5.4	4.4.2
Frequency response	3.5.5	4.4.3
Distortion	3.5.6	4.4.4
Noise and microphonics	3.5.7	4.4.5
Transients	3.2	MIL-I-983
Output voltage regulation	3.5.8	4.4.6
Stability	3.5.9	4.4.7

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4.3.3 Sampling for group C tests.- Group C tests will be required by the bureau or agency concerned when the basic design of the equipment or the material of a vital part has been changed. One complete equipment shall be selected and subjected to each of the group C tests in the order listed. The results of each test shall be compared with specification requirements. In the event of failure to conform to this specification for any group C test, the contractor shall correct the cause of failure on future production units and repair the deficiency in all equipments produced on the contract or order.

Group C tests		
Radio frequency noise interference	3.2	MIL-I-983
Temperature and humidity	3.2	MIL-I-983
Enclosure	3.2	MIL-I-983
Shock and vibration	3.2	MIL-I-983

4.4 Test procedures.-

4.4.1 Operating test.- The operating test shall consist of ascertaining compliance with 3.4.1 to 3.4.4 inclusive, using sound powered telephones and loudspeakers or simulating these units with suitable electrical loads.

4.4.2 General design characteristics.- Tests for determination of compliance with 3.5.1 to 3.5.4 inclusive shall be made with suitable high quality commercial meters.

4.4.3 Frequency response.- At rated gain with a constant sine-wave voltage input signal such that the amplifier does not overload at any frequency between 200 and 4,000 c.p.s., the output voltage at the output terminals for one loudspeaker across a resistive load of such a value as to draw rated output power at 70 volts, shall be as specified in 3.5.5.

4.4.4 Distortion.- At rated gain and with a sine-wave voltage having a frequency of 1,000 c.p.s. applied to the input terminals of such value as to draw rated output power (70 volts across a resistive load of 650 ohms) at the output terminals of the loudspeaker, the total harmonic distortion of the output voltage shall not exceed the value specified in 3.5.6.

4.4.5 Noise and microphonics.- At full gain and with an 800-ohm resistor connected to the amplifier input terminals, the output voltage at the output terminals of the loudspeaker across a resistive load of such a value as to draw rated output power at 70 volts when the input is 25 ± 2 millivolts r.m.s. shall not exceed the value specified in 3.5.7. The output voltage at one pair of the sound powered telephone output terminals across a resistive load of 600 ohms shall not exceed the value specified in 3.5.7. Both measurements shall be made during the vibration test, and shall be made with an unweighted full range voltmeter. When acceptance tests do not require a vibration test, these measurements shall be made under ambient conditions.

4.4.6 Output voltage regulation.- At rated gain and with a steady state 1,000 c.p.s. sine-wave signal of 25 ± 2 millivolts r.m.s. applied to the amplifier input terminals, a variation of amplifier output load, at the output terminals of the loudspeaker, from full load (that resistive load which will draw rated output power at 70 volts) to no load (open circuit) shall not exceed the value specified in 3.5.8.

4.4.7 Stability.- The amplifier shall be operated at full gain with the amplifier input either short or open circuited and with the amplifier output either short or open circuited to determine conformance with 3.5.9.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation and for storage of shipboard repair parts.-

5.1.1 Basic equipment or item.-

5.1.1.1 Preservation and packaging.- Preservation and packaging shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation and may conform to the suppliers commercial practice when such meets these requirements.

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5.1.1.2 Packing. - Packing shall be accomplished in a manner which will insure acceptance by common carrier at the lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may conform to the suppliers commercial practice when such meets these requirements.

5.1.1.3 Marking. - Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, contractor's name and destination.

5.1.2 Shipboard repair parts. - The shipboard repair parts shall be preserved and packaged level A; packed level C and marked levels A and C respectively in accordance with MIL-E-17555.

(5.2 Domestic shipment and storage or overseas shipment. - Where special requirements not covered by 5.1 are required, the levels of preservation, packaging, packing and marking for shipment shall be specified by the procuring activity (see 6.2).

5.2.1 The following provides various levels of protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made (see 6.2).

5.2.1.1 Preservation, packaging, packing and marking. - The equipment and accessories, repair parts and technical manuals shall be preserved and packaged levels A or C; packed level A or B as specified (see 6.2) and marked in accordance with MIL-E-17555.)

6. NOTES

6.1 Intended use. - The sound powered telephone amplifiers covered by this specification are intended for use in sound powered telephone circuits, to amplify audio frequency signals sufficiently to drive a group of telephones and loudspeakers.

6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Quantity of items required.
(c) Quantity of manuals required (see MIL-M-15071).
(d) Supply demand control point for stock repair parts (see MIL-E-17362).
(e) Preservation and packaging, packing, and marking if other than as specified in 5.1 (see 5.2).

6.3 Preproduction. -

6.3.1 Invitations for bids should provide that the Government reserves the right to waive the requirement for preproduction samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

6.3.2 Procurements for the Bureau of Ships shall provide for delivery of a preproduction sample to a Government laboratory for tests after successful completion of the tests to be made by the supplier.

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Preparing activity:
Navy - Ships
(Project 5805-N033Sh)

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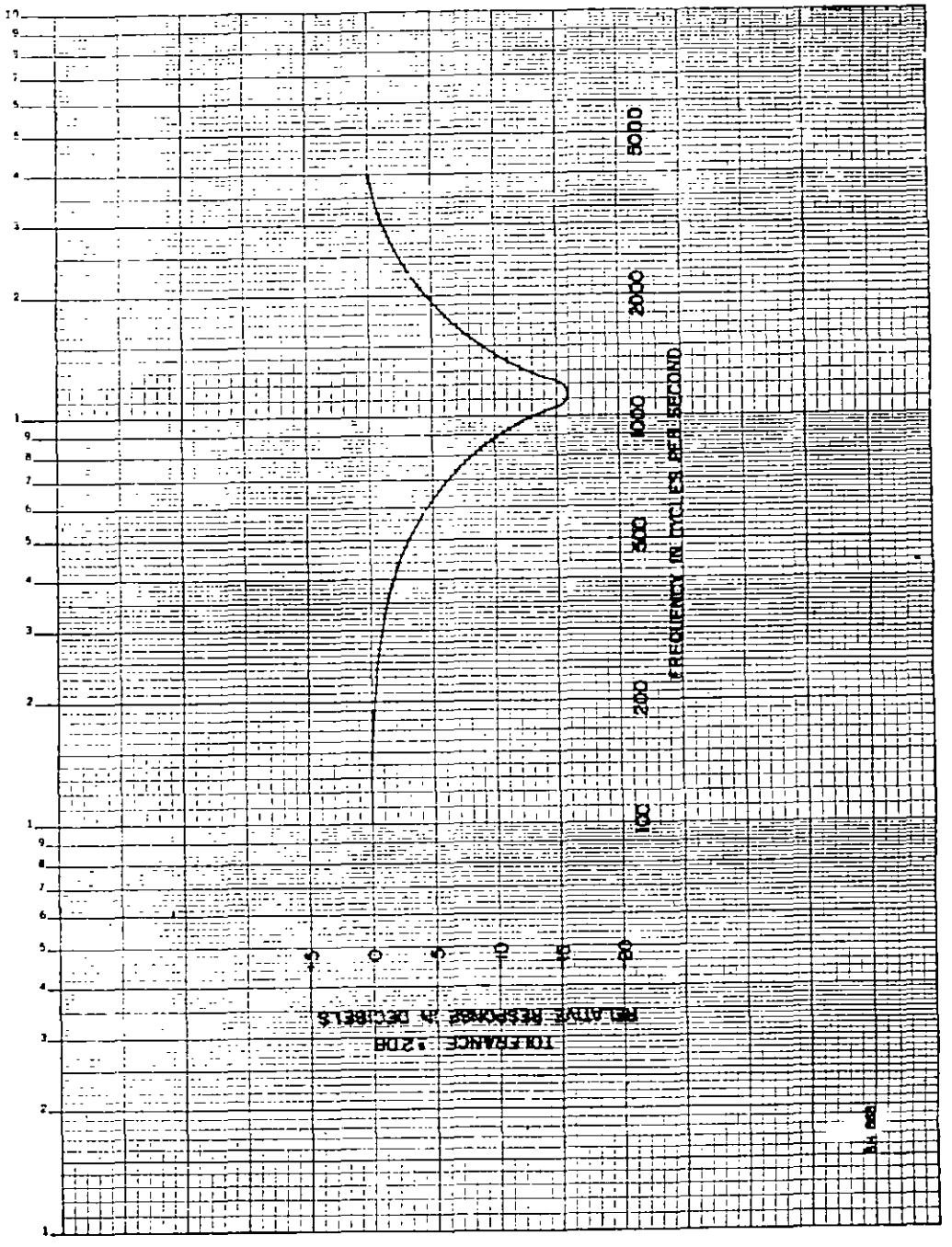


Figure 1 - Frequency response limits.

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SPECIFICATION ANALYSIS SHEET

Form Approved
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INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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